



Modelling the potential alternative pathways for the transition to clean cooking for the Residential sector in Ethiopia



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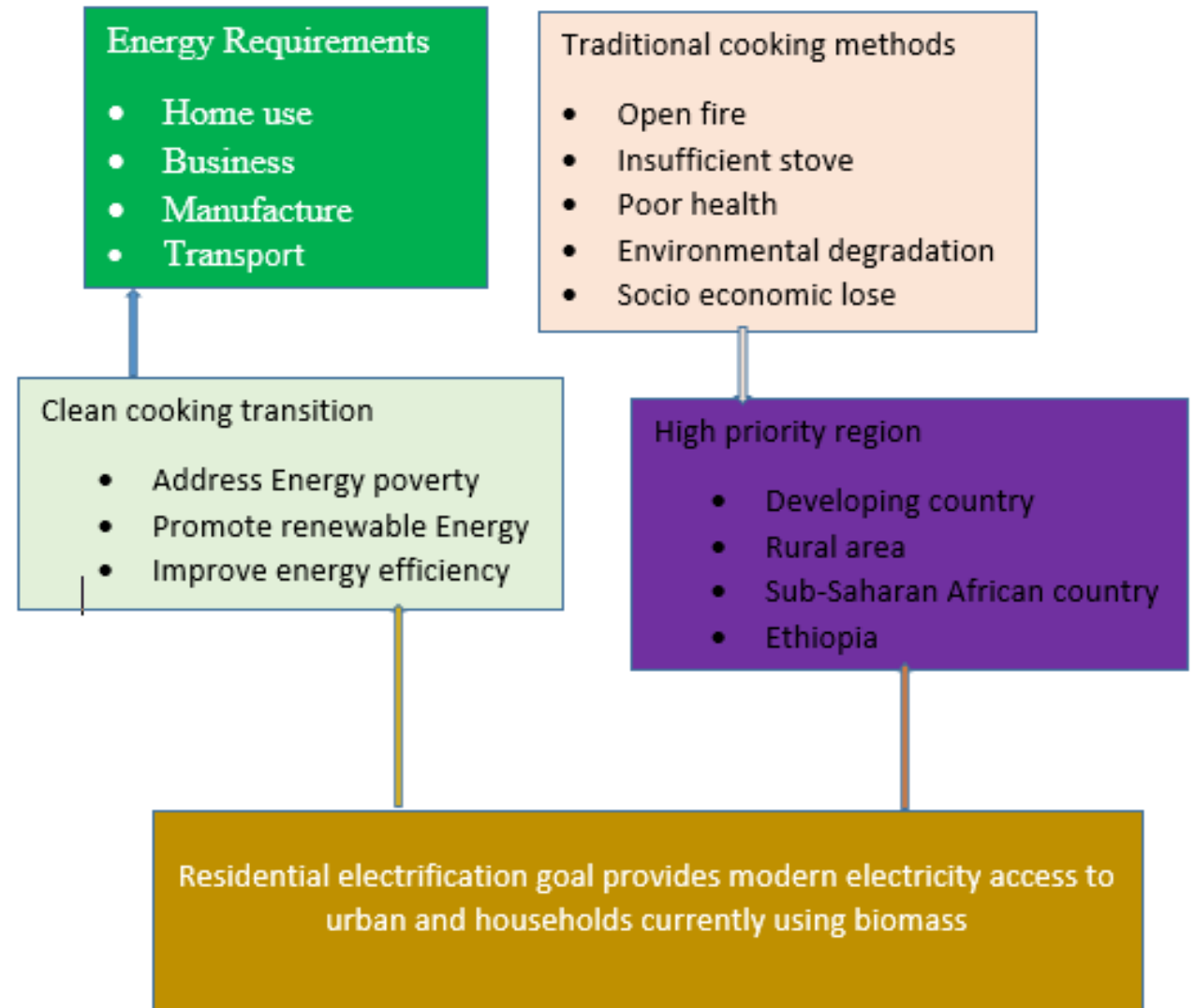
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Energy Modelling Platform Ethiopia (EMP-E)

2026

Outlines

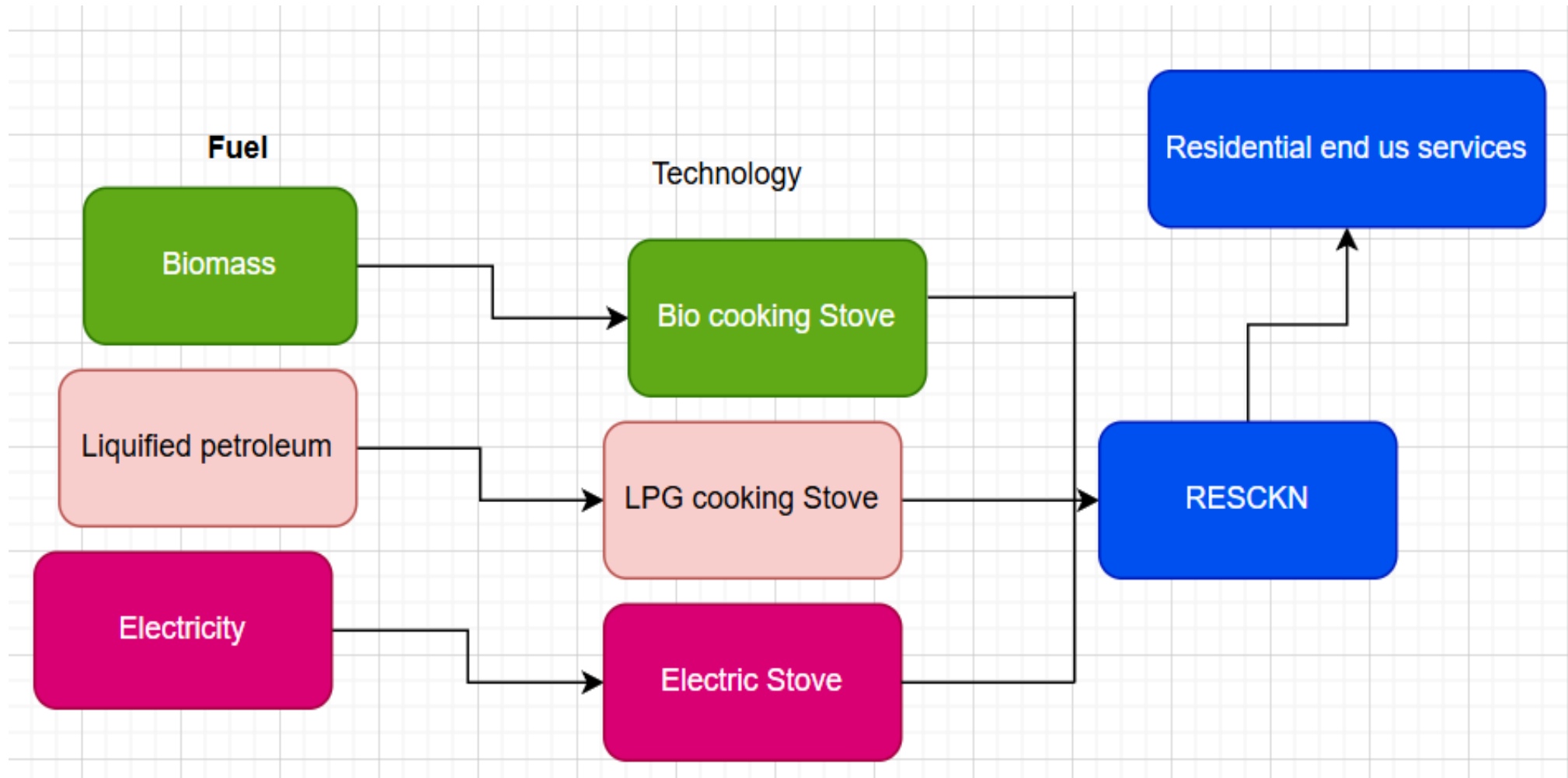
- Context, challenges & Research Question
- Reference Energy System
- Scenarios
- Results
- Conclusions, Policy insights, & Future Work



[ETHIOPIAN] Reference Energy system for residential cooking demand

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- ✓ In order to incorporate the clean cooking technologies in the OSeMOSYS model, certain key elements related to the clean cooking technologies have to be considered.



Model-Based Assessment of Clean Cooking Transition

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Main Challenges Investigated

- Heavy dependence on traditional biomass for residential cooking
- Unrealistically rapid technology shifts in least-cost models
- Affordability and accessibility constraints for clean cooking fuels

Main Research Questions

- ❖ How does residential cooking energy transition under different policy scenarios?
- ❖ What role does LPG play as a transition fuel toward full electrification?
- ❖ How realistic is a rapid shift to electric cooking in the Ethiopian context?

Main Findings (per Cooking Technology)

- ❑ **Traditional Biomass:** Gradually declines and is fully phased out after 2037 in clean cooking scenarios
- ❑ **LPG:** Increases in the medium term due to affordability and accessibility, then declines after 2050
- ❑ **Electric Cooking:** Becomes the dominant technology by 2050 under strong policy support

Energy System Model the following scenarios were investigated:

Scenario Label	Scenario Description	Key Assumptions
BAU	Biomass continues to supply most residential cooking energy until 2037.	✚ Rural cooking dominated by traditional biomass stoves. ✚ Minimal adoption of clean cooking technologies. ✚ No policy restrictions on traditional biomass use.
Clean Cooking / Electrification	Biomass use declines as LPG and electric cooking expand, with full electrification by 2050.	➤ Biomass reduced and LPG increases from 2026–2050 ➤ Electricity becomes the main cooking fuel by 2050 ➤ LPG serves as a transition fuel and is replaced by electric stoves after 2050

Business-as-Usual Scenario modeling

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- In the Business-as-Usual scenario generally refers to a situation where existing trends and policies continue without much change.
- As a result of these predictions, the biomass sector will continue to provide the majority of residential cooking energy.
- In our case electricity will continue to be used dominantly for cooking after 2037 .

BAU Total Production By Technology By Mode

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Production by technology by mode filtered by demand

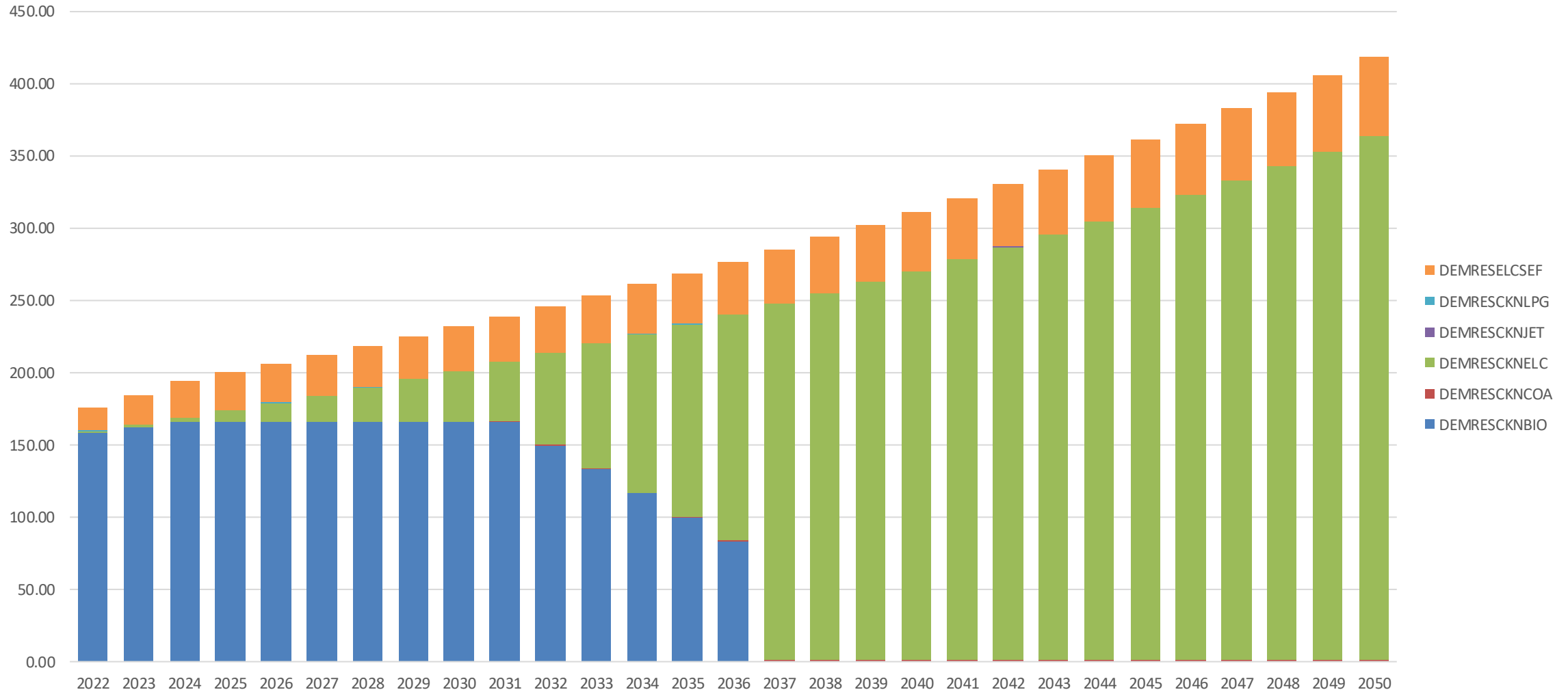


Figure 2: Total Capacity By Technology

BAU Total Production By Technology By Mode

From previous slide Figure 2 we analysed that;

- This abrupt shift may not reflect real-world conditions, where transitions are usually gradual.
- Technology adoption is influenced by consumer behavior, government subsidies, and cost changes.
- The increase in LPG use is driven by improved accessibility and affordability.
- There is a clear shift from traditional biomass (e.g., firewood) to cleaner alternatives.
- The model indicates that no traditional biomass cookstoves are used after 2037.

Total Annual Technology Activity By Mode for base scenario

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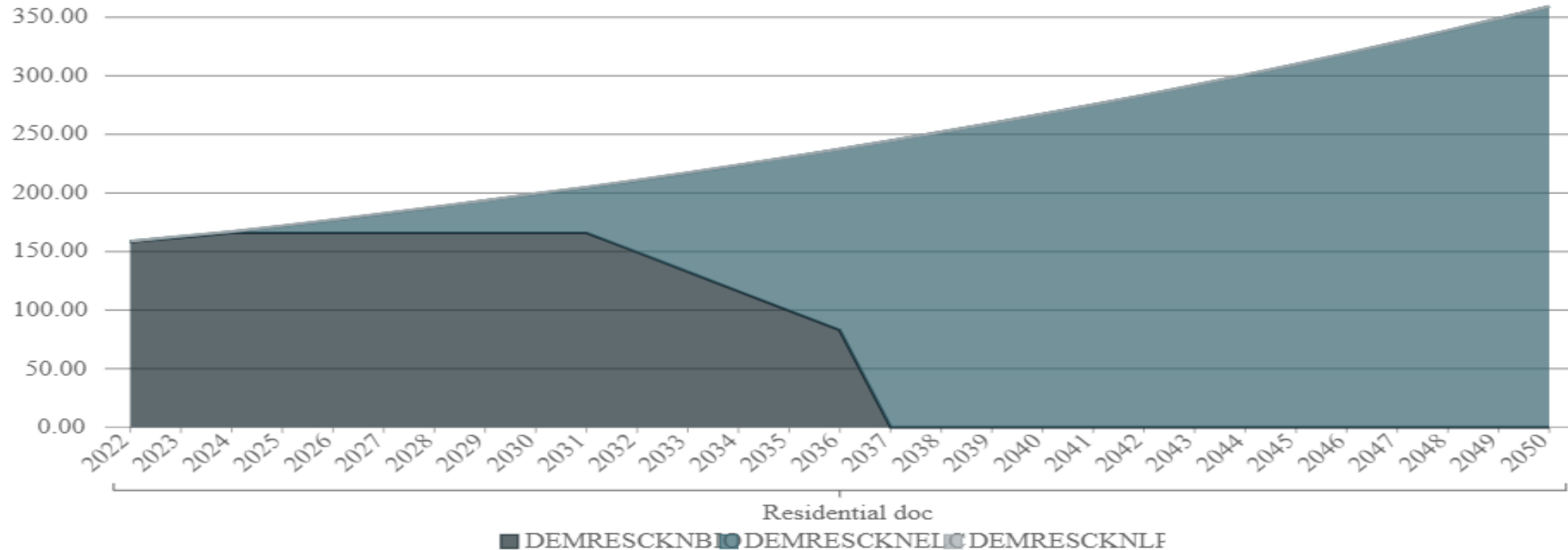


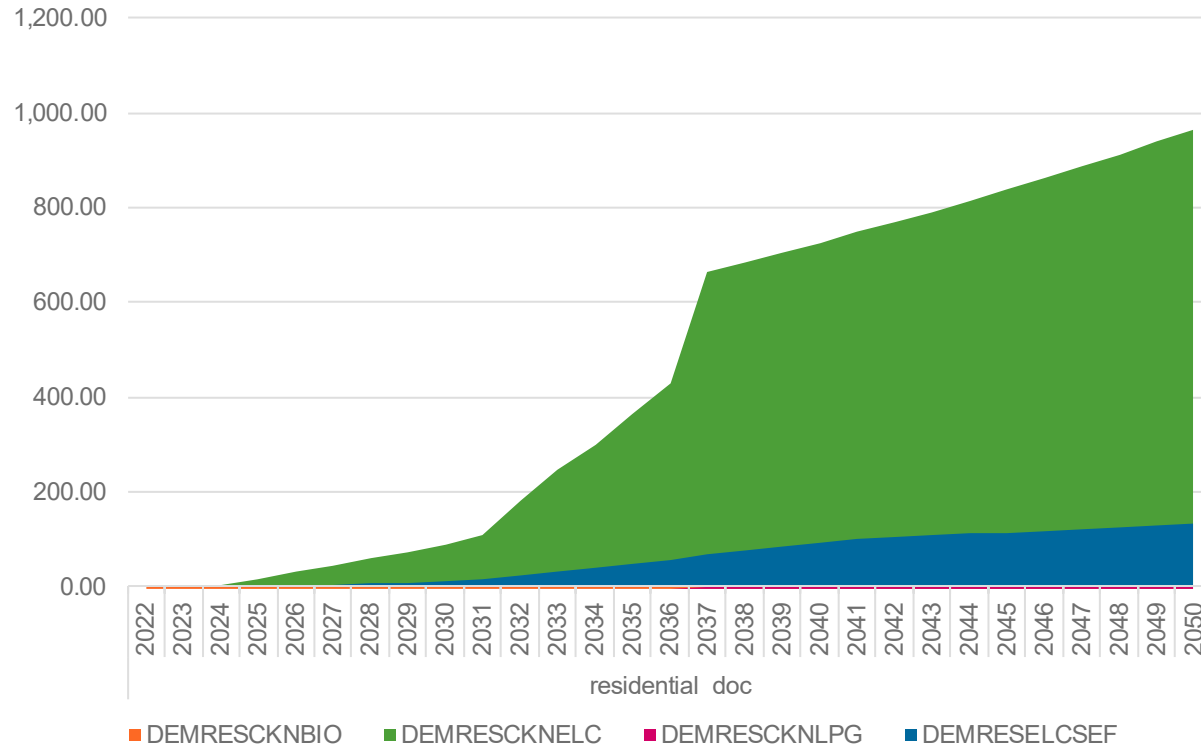
Figure 3: Total annual Technology activity by mode

- ✓ Total Technology annual activity upper Limit parameter is provided as a ratio of the RESCKN demand, allowing for the introduction of clean cooking technologies and the eventual phase-out of non-conventional cooking technologies.

Annualized investment cost(M USD)

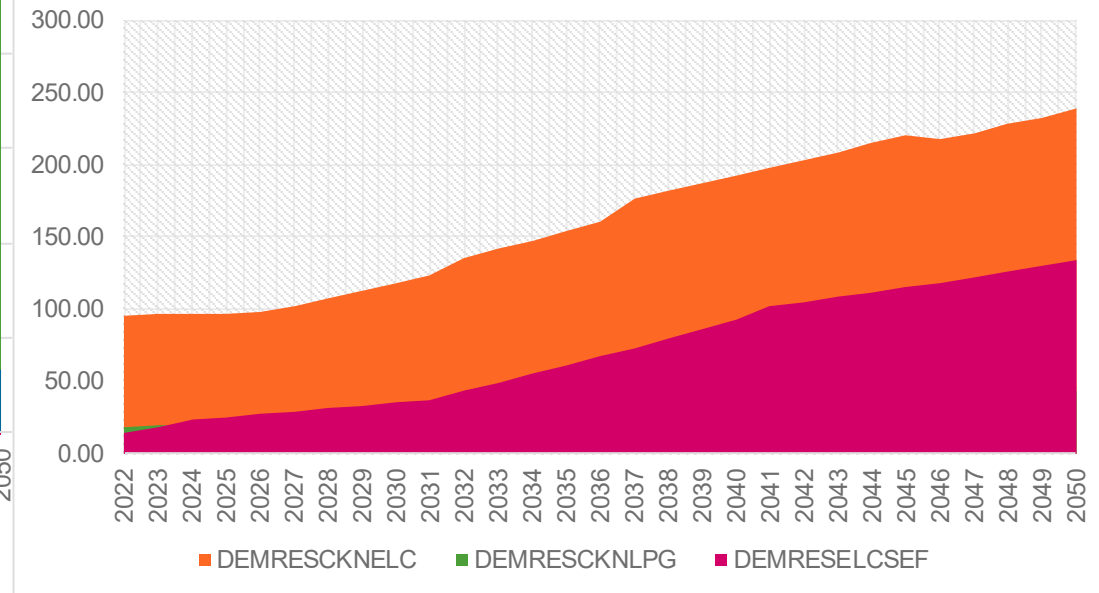
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Annualised investment cost



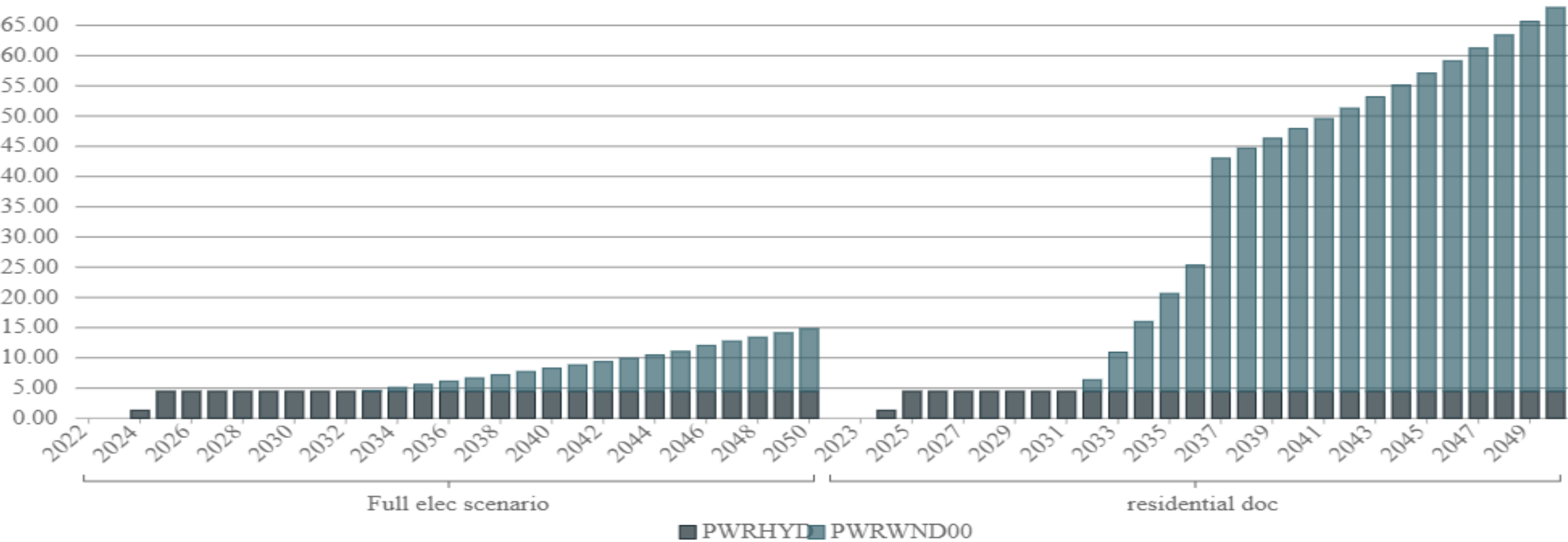
BAU Scenario

Annualised investment cost for clean cooking access



Clean cooking access scenario

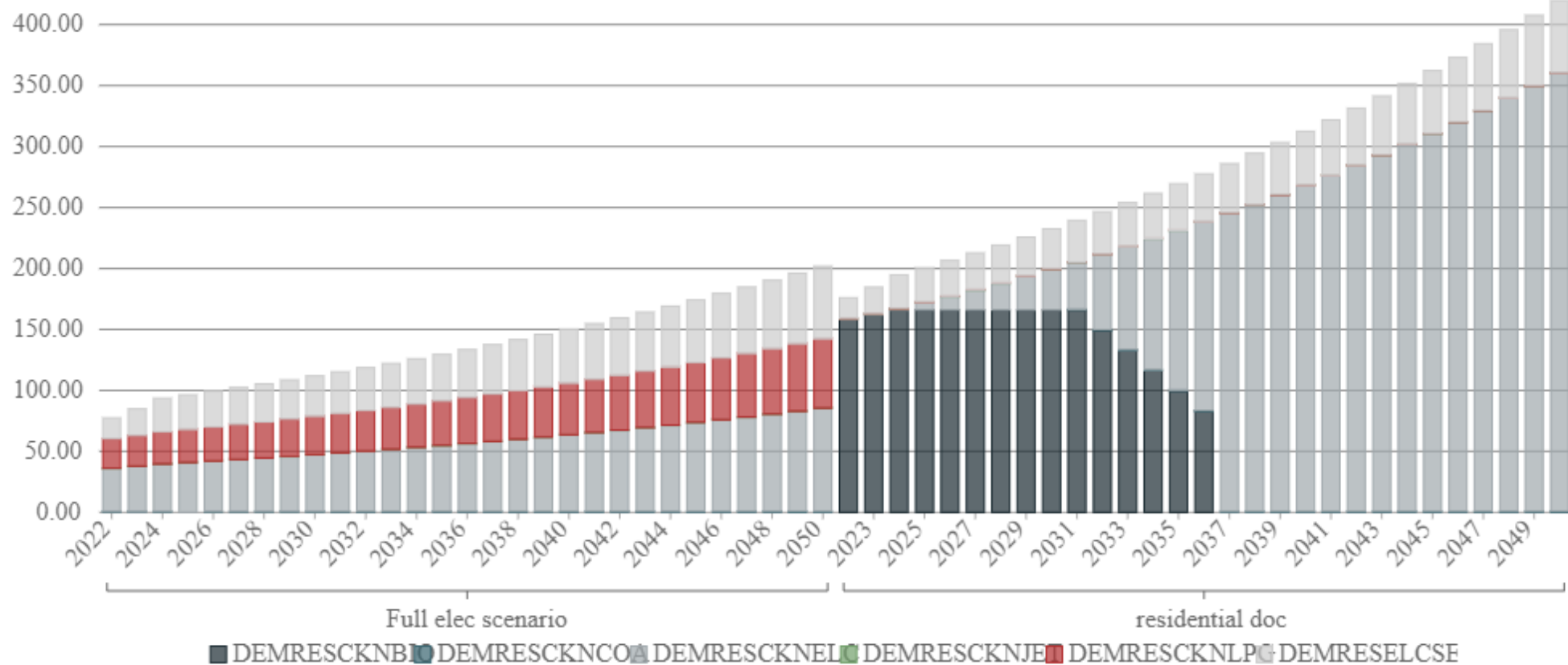
Accumulated new capacity(PJ)



BAU Scenario

Clean cooking access scenario

Production by Technology by Mode



From the previous slide, the Production by Technology by mode figure suggests that;

- There is a noticeable shift in the shares of cooking technologies under the fully electric cooking scenarios.
- Biomass stoves are the dominant cooking technology for the residential sector in the reference scenario, but their share decreases over time in the transition scenarios as electric and LPG stoves become cost-competitive and available.
- By the end of the model horizon, electric stoves meet the majority of the residential cooking sector's energy demand, implying the complete or near-complete phase-out of traditional stoves.

- OSeMOSYS was used to assess the transition from biomass and fossil fuel stoves to electric cooking.
- A reference scenario was compared with electric cooking transition scenarios.
- Electric stoves gradually replace biomass and LPG technologies.
- Assumptions include improved electricity access and grid reliability.
- Electric stove efficiency improves over time.
- Conventional cooking technologies are constrained to support clean cooking goals.

Conclusions

- ❖ The model shows a strong shift from traditional biomass to clean cooking technologies.
- ❖ LPG plays a key transitional role before large-scale electric cooking adoption.
- ❖ Rapid electrification projected by the model may be optimistic compared to real-world adoption dynamics.

Policy Insights

- Promote LPG as a short- to medium-term transition fuel.
- Support electric cooking through targeted subsidies and grid reliability improvements.
- Gradually restrict traditional biomass use while aligning policies with consumer behavior and affordability.

Acknowledgements

- ✓ Climate Compatible Growth
- ✓ World Bank Group
- ✓ 2050 Pathways Platform
- ✓ British Embassy of Buenos Aires
- ✓ Jimma university
- ✓ This material has been produced with support from the Climate Compatible Growth (CCG) programme. CCG is funded by UK aid from the UK government.



We would like to express a special thanks to our trainer Joshua odour, Dr. Emiyamirew, Birhanu and Kifle for their invaluable guidance throughout this training.

We also extend our gratitude to all the members of OSeMOSYS and the EMP-LAC 2025 team for their collaboration and significant contributions.

Thank You